Comparing Traditional Machine Learning and Deep Learning for Multiclass Air-Quality Classification Using Beijing Monitoring Data

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Introduction to machine learning,
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16/10/2026.

GitHub Repository:
https://github.com/Akinloye3264/beijing-air-quality-classification

Google Drive Folder:
ML_Final_Project_Beijing_Air_Quality

Video Presentation: https://youtu.be/kjr3i276fxQ

ABSTRACT

Classification of pollution severity can be considered an environmentally and health-related issue of considerable importance. This project aims to analyze the performance of classical machine learning and deep learning algorithms in classification of hourly air quality levels into four PM2.5 severity categories. The Beijing Multi-Site Air Quality dataset comprised 420,768 samples obtained from 12 air quality monitoring stations during March 2013 through February 2017. After removing the records with missing PM2.5 targets, 412,029 observations were left. Then, the dataset was split according to the chronological order into 70% training, 15% validation and 15% test subsets to ensure that future observations would not affect the process of model training. In eight experiments, Logistic Regression, Random Forest, Histogram Gradient Boosting and several feedforward neural networks were used. As a metric for the models comparison, macro-F1 was chosen due to moderate imbalance of the target classes. Histogram Gradient Boosting turned out to yield the best test results with accuracy, macro-F1 and macro ROC-AUC equal to 0.8580, 0.8226 and 0.9731, respectively. The deep neural network optimization yielded the following test performance metrics: 0.8430, 0.8067 and 0.9689, respectively. PM10, carbon monoxide and dew point proved to be the most impactful predictors. The error analysis revealed that misclassification tended to happen around the threshold values for PM2.5 used to obtain the target classes. The results show that a classical boosting model can outperform a feedforward neural network on the structured environmental data.

Index Terms — Air pollution, PM2.5, climate change, air-quality classification, machine learning, deep learning, gradient boosting, neural networks.

I. INTRODUCTION

It is well-known that air pollution is one of the most relevant environmental issues that negatively impact people's health, economic performance, ecosystem functioning and sustainability. It causes a variety of diseases, such as strokes, ischaemic heart disease, chronic obstructive pulmonary disease, asthma, and cancer [1]. Moreover, according to the research into the global burden of disease,

long-term exposure to ambient PM2.5 is associated with higher mortality rates, particularly, in the countries with high levels of pollution [2]. Such facts prove that air pollution control and analysis should be carried out reliably and effectively. Air pollution and climate change have much in common due to the same emission sources, such as fossil-fuel combustion, transport activity, industries, waste burning, and energy consumption in households. Thus, measures that reduce emissions of greenhouse gases and pollutants could bring environmental and health benefits to the community. However, before intervention it is necessary to know the time and location of air pollution.
I would like to emphasize that my personal mission deals with climate change and environmental sustainability. The reason why I chose air-quality classification for this project is the problem of air pollution in African cities and poor access to air pollution information there. According to research, sub-Saharan Africa has low air quality monitoring and fewer epidemiological studies than other continents [3]. Moreover, health and economic effects of air pollution in Africa are growing; however, the poor monitoring infrastructure does not allow assessing air pollution exposure and developing relevant measures [4]. Low-cost sensor studies have already proved the possibility of air pollution monitoring in Africa despite difficulties with maintenance, power supply, Internet, etc. [5].

In this case, the current project does not assume the use of a machine-learning model to reduce the level of pollution. However, classification models could classify sensor measurements into different severity categories. Such an approach might prove useful for environmental monitoring, notifying the population about risks, reducing exposure, and making evidence-based policy decisions. The current dataset might be helpful as an extensive open-source dataset for further research purposes. It should be noted that the models developed on the data from Beijing cannot be used in African cities because they require validation in the area of application. Nevertheless, the approach taken in the current project might become a guide for future research when using air quality data from African cities. The purpose of the study is the comparison of traditional machine learning approaches and deep learning in solving an air quality classification problem. PM2.5 values were classified into four project-specific categories: Low, Elevated, High, and Very High. PM2.5 values were removed from the list of predictors as they determine the target variable directly. Other predictors include other types of pollutants, meteorological parameters, characteristics of monitoring stations, direction of the wind, temporal predictors, and predictors of missing values.

Five traditional machine-learning and three neural-network experiments were evaluated. The models included Logistic Regression, Random Forest, Histogram Gradient Boosting and multilayer neural networks. Performance was assessed using accuracy, macro precision, macro recall, macro-F1, weighted F1 and multiclass ROC-AUC.

The study was guided by four research questions:

1. How do traditional machine-learning and deep-learning models compare when evaluated on future chronological observations?

2. Does class weighting improve performance across the four air-quality classes?

3. Which predictor variables contribute most strongly to the selected traditional model?

4. Are classification errors concentrated near the PM2.5 boundaries used to construct the target classes?

Beyond identifying the highest-performing model, the project aims to explain how the models behave. Learning curves, confusion matrices, ROC and precision–recall curves, permutation feature importance, and boundary-based error analysis were used to examine generalization, class-level weaknesses, and possible reasons for incorrect predictions.

II. LITERATURE REVIEW

A. Air Pollution, Health, and Climate

PM2.5 is one of the most important air pollutants because it can bypass the respiratory system and reach the blood stream [1], [2]. This issue is particularly challenging to evaluate in sub-Saharan Africa where the lack of monitoring and epidemiological data can lead to underestimation of health effects [3]. Wright et al. [4] also found increased health and economic impact with unmet monitoring coverage. Similarly to climate change, air pollution has common anthropogenic sources (fossil fuel combustion, transport, industry, biomass burning). Cutting the emissions from those sources can then bring the benefits of improving health and climate. Awokola et al. [5] proved that PM2.5 long-term monitoring with low-cost sensors can be performed in Africa, even if electricity supply, connectivity, maintenance and security are still real challenges. It's thus essential to have a reliable infrastructure for efficient data-based analysis

B. Traditional Machine Learning for Air-Quality Analysis

Machine learning algorithms are still relevant in cases where the data are structured. Logistic Regression can provide a linear baseline with interpretability, but modeling pollution levels with weather, location, time of day, may be more complex than a linear model can cope with. Random Forest leverages the strength of many randomized decision trees in modeling nonlinear data and is less susceptible to over-fitting of individual trees [6]. Gradient boosting builds predictive models with trees sequentially, where subsequent trees can use the residuals of previous trees to improve the fit [7]. Histogram-optimized boosting solves the scalability problem by clustering continuous feature values into bins to lessen the cost of searching for the best split. Ameer et al. [8] benchmarked different machine-learning models and showed ensemble machine-learning models as promising but also with their caveats based on the data set and evaluation strategy. Harishkumar et al. [9] also identified meteorological variables as predictive of PM2.5. Morapedi and Obagbuwa [10] showed that machine learning can help predict PM2.5 levels in South African cities, but they also pointed out problems with limited samples and data that doesn’t fully represent local conditions. Often, in machine learning, randomly assigning data to train and test sets would place temporally correlated observations in both halves and would give overly optimistic estimates of how well the model performs on future data. Additionally, predicting accuracy will mask the variable performance of pollutant categories with low concentrations.

C. Deep Learning for Air-Quality Analysis

Deep-learning algorithms can learn complex, nonlinear relationships from large datasets. In air-quality applications, pollutant data has been integrated with meteorological data and recurrent, convolutional and hybrid neural network structures have been implemented for time-dependent prediction [11]. Bekkar et al. [12] demonstrated the ability of deep networks, although the authors identified issues related to data availability, pre-processing, choice of algorithm architecture and computational cost. Iskandaryan et al. [13] found that PM2.5 was a popular choice as a prediction target and that meteorological and spatiotemporal features were commonly used in conjunction with other pollutants. Deep learning does not necessarily outperform traditional algorithms when working with tabular data, however; experimental results are influenced by algorithm architecture, regularisation, training procedure and optimisation, and deep learning models require additional time and resources to tune. As this study aims to classify the current state from simultaneously-measured data, while prediction models would work with sequence data, feedforward neural network structures are a sensible baseline to compare with traditional tabular algorithms; sequence modelling will be considered in future research into pollutant prediction.

D. Research Gap

Previous studies demonstrate the value of ensemble learning and neural networks for air-quality analysis, but comparisons are complicated by differences in datasets, targets, preprocessing and evaluation procedures. Some studies also rely mainly on accuracy and provide limited analysis of minority classes. This project addresses these limitations through chronological data partitioning, macro-averaged evaluation, TensorBoard experiment tracking and boundary-based error analysis. The design compares traditional and deep-learning models while examining both predictive performance and the causes of misclassification

III. METHODOLOGY

A. Dataset and Study Design

This study used the Beijing Multi-Site Air Quality dataset from the UCI Machine Learning Repository [14], which accompanies research by Zhang et al. [15]. It contains 420,768 hourly observations from 12 Beijing monitoring stations between March 2013 and February 2017. The variables include six air pollutants: PM2.5, PM10, SO_2, NO_2, CO and O_3, together with meteorological, station and time information. PM2.5 was used to construct the target and excluded from the predictors to prevent leakage. Removing observations without a PM2.5 value left 412,029 usable records. All remaining preprocessing statistics were calculated only from the training partition.

B. Target Variable Construction

PM2.5 measurements were divided into four classes: Low (≤35.4), Elevated (>35.4–55.4), High (>55.4–150.4) and Very High (>150.4). The respective class proportions were 37.34%, 12.92%, 34.52% and 15.21%. Because the distribution was moderately imbalanced, class-weighted models were evaluated and macro-averaged metrics were used.

C. Chronological Data Partitioning

Data was separated in a chronological order, since adjacent hourly data points are temporally correlated. Randomly separating the data could have resulted in similar times appearing in both the training and evaluation data, leading to an inflated estimate of future performance. The data was 70% training, 15% validation, and 15% test, as shown in Table I. The validation set was used for model selection and hyperparameter tuning, with the test data solely used for the final models.

TABLE I

CHRONOLOGICAL DATA PARTITIONING

PARTITION	OBSERVATIONS	PERCENTAGE	START DATE	END DATE
Training	288,258	69.96%	1 Mar, 2013 00:00	18 Dec, 2015 05:00
Validation	61,785	15.00%	18 Dec, 2015 06:00	24 Jul, 2016 05:00
Test	61,986	15.04%	24 Jul, 2016 06:00	28 Feb, 2017 23:00
Total	412, 029	100.00%	1 Mar, 2013	28 Feb, 2017

D. Data Preprocessing and Feature Engineering

Missing predictors were imputed using station-specific training medians, followed by overall training medians where necessary. Missingness indicators were retained as additional features. Date and time columns were combined into a timestamp, while hour, month and day of week were represented using sine and cosine transformations. Station and wind direction were one-hot encoded, producing 55 predictors. Numerical variables were standardized for Logistic Regression and the neural networks using training-derived statistics. Scaling was unnecessary for the tree-based models.

E. Experimental Models

Eight experiments were conducted. EXP-01 and EXP-02 were unweighted and class-weighted Logistic Regression models. EXP-03 was a 150-tree Random Forest, while EXP-04 added regularization and class weighting. EXP-05 used class-weighted Histogram Gradient Boosting with 200 iterations, a learning rate of 0.08 and L2 regularization. Random Forest and gradient boosting were included because they can represent nonlinear interactions in structured data [6], [7]. EXP-06

was a baseline multilayer perceptron with 128- and 64-unit hidden layers. EXP-07 introduced a larger class-weighted architecture, batch normalization, dropout and early stopping. EXP-08 added Swish activations, regularization and learning-rate scheduling. The neural networks used a TensorFlow `tf.data` pipeline with a batch size of 512. Training data were shuffled, whereas validation and test data retained their order. TensorBoard recorded accuracy, loss and learning-rate behaviour.

F. Evaluation Procedure

Models were selected primarily using validation macro-F1, which gives each class equal importance and is therefore appropriate for the imbalanced target. Additional measures included accuracy, macro precision, macro recall, weighted F1 and one-versus-rest macro ROC-AUC. Per-class metrics, confusion matrices, ROC curves and precision-recall curves were also produced. EXP-05 and EXP-08 were selected as the strongest traditional and deep-learning models and evaluated once on the test set. A chronological learning curve assessed EXP-05 as the training period increased, while permutation importance measured the validation macro-F1 reduction caused by shuffling each predictor. Finally, boundary analysis grouped test observations by distance from the nearest PM2.5 threshold to determine whether errors were concentrated near class boundaries.

G. Reproducibility

A fixed random seed was used wherever supported. The fitted models, preprocessing objects, feature definitions and model manifest were saved as reusable artifacts. Reload tests confirmed that the stored traditional model, neural network and preprocessing pipeline could be loaded successfully and used to reproduce predictions. Source code, dependencies and generated figures were also organized in the public project repository.

TABLE II
VALIDATION PERFORMANCE OF THE EIGHT EXPERIMENTS

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Macro ROC_AUC	Time (s)
EXP- 01 Logistic Regression	0.7917	0.7507	0.6957	0.7076	0.7702	0.9367	2.10
EXP-02 Weighted Logistic Regression	0.7592	0.7555	0.7652	0.7406	0.7727	0.9376	2.59

EXP-03 Random Forest	0.8260	0.8073	0.7456	0.7642	0.8129	0.9601	2.03
EXP-04 Weighted Random Forest	0.8244	0.7985	0.8015	0.7974	0.8289	0.9590	6.16
EXP-05 Histogram Gradient Boosting	0.8233	0.8025	0.8240	0.8041	0.8321	0.9646	6.19
EXP-06 Baseline MLP	0.8338	0.8003	0.7820	0.7899	0.8298	0.9598	52.68
EXP-07 Regularized DNN	0.8240	0.8043	0.8272	0.8060	0.8334	0.9642	28.89
EXP- 08 Optimized DNN	**0.8335**	**0.8058**	**0.8335**	**0.8125**	**0.8410**	**0.9669**	**33.07**

IV. RESULTS

A. Validation Results

As shown in Table II, class weighting improved macro-F1 for Logistic Regression and Random Forest, although it did not always improve accuracy. EXP-05 produced the strongest traditional-model macro-F1 at 0.8041. EXP-08 achieved the highest overall validation macro-F1 at 0.8125 and was selected as the final deep-learning model. The neural models required substantially more training time than the traditional alternatives.

B. Final Test Results

After model selection using the validation set, EXP-05 and EXP-08 were evaluated on the untouched chronological test partition. Table III presents their final test performance.

TABLE III

FINAL TEST PERFORMANCE OF THE SELECTED MODELS

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Macro ROC_AUC
EXP-05 Histogram Gradient	0.8580	0.8125	0.8482	0.8226	0.8646	0.9731

Boosting						
EXP-08 Optimized DNN	0.8430	0.7975	0.8360	0.8067	0.8508	0.9689

EXP-05 achieved the strongest final test performance, with 0.8580 accuracy, 0.8226 macro-F1 and 0.9731 macro ROC-AUC. EXP-08 achieved 0.8430 accuracy, 0.8067 macro-F1 and 0.9689 macro ROC-AUC. Therefore, the boosting model outperformed the optimized neural network across all reported test measures. EXP-05's macro-F1 increased from 0.8041 on validation data to 0.8226 on test data, whereas EXP-08 decreased slightly from 0.8125 to 0.8067. Both models nevertheless retained macro ROC-AUC values above 0.96.

C. Class-Level Performance

Both models performed most strongly on the Low and Very High classes. Elevated was the most difficult category, with F1-scores of 0.6072 for EXP-05 and 0.5858 for EXP-08. Most errors occurred between neighbouring categories, particularly Elevated and High. Direct confusion between Low and Very High was rare, indicating that both models generally preserved the ordering of pollution severity.

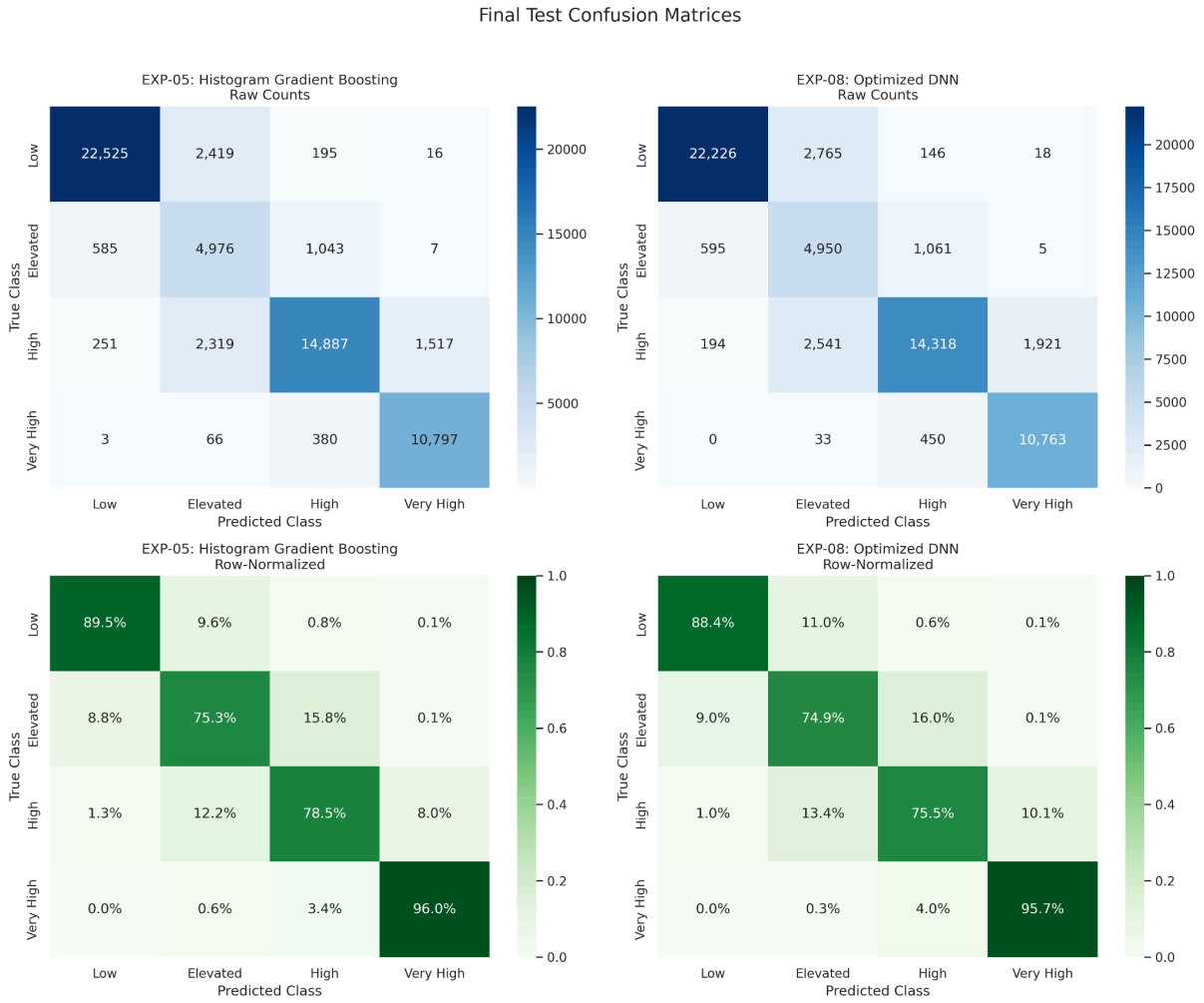


Fig. 1. Raw-count and row-normalized confusion matrices for EXP-05 and EXP-08 on the final chronological test set.

D. Learning Curve and Feature Importance

The chronological learning curve of EXP-05 is presented in Figure 2, and the training histories of EXP-08 are provided. The macro-F1 scores for EXP-05 are 0.7343 (20% of training set) and 0.8041 (full training set). In the same time frame, training macro-F1 dropped from 0.8778 to 0.8382. The decrease in the gap between training and validation performance indicates improved generalization with the increase in chronological observations. The training accuracy was consistently higher in EXP-08 and the validation accuracy was kept near the maximum value. The loss value for validation set was minimum at epoch 13 which was the result of early stopping and the best model weights were saved at that time. The permutation importance of EXP-05 revealed that the most important predictor is PM10, with a mean macro-F1 decrease of 0.3720 after shuffling. CO and dew point were then next, with decreases of 0.1780 and 0.1318, respectively. O_3, temperature and seasonal features made smaller positive contributions, while rainfall and several station indicators had little individual importance.

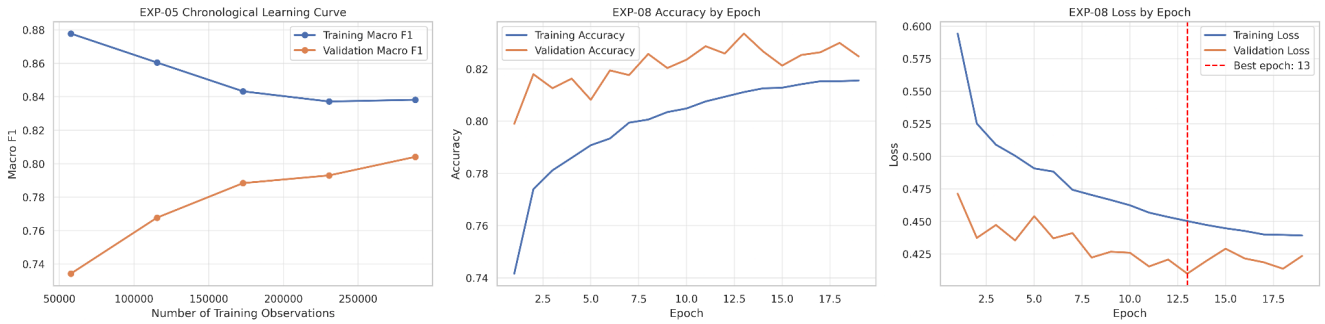


Fig. 2. Chronological learning curve for EXP-05 and accuracy and loss histories for EXP-08.

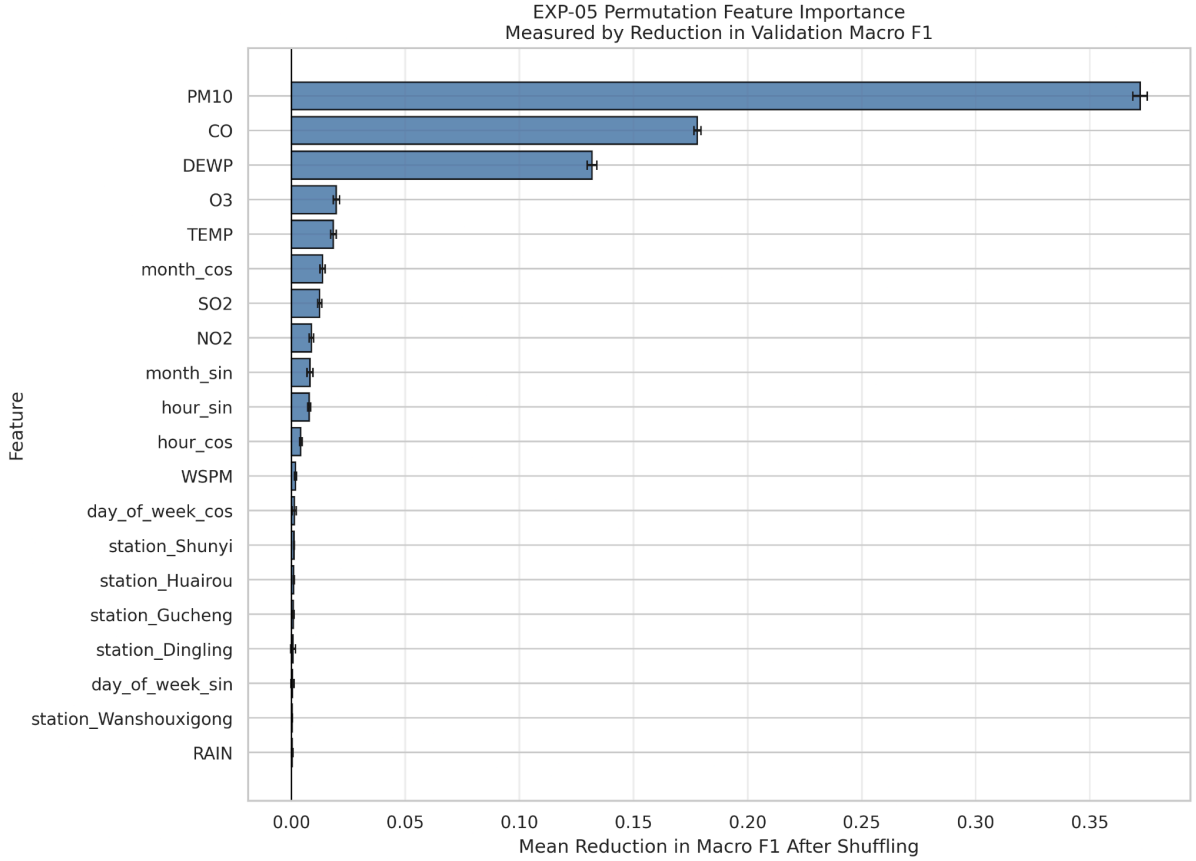


Fig. 3. Permutation feature importance for EXP-05 measured by reduction in validation macro-F1

E. Boundary Error Analysis

Figure 4. Model error rates by distance from the nearest PM 2. In both models, the error rate was lowest within two units of a boundary. For any observation within two units of a boundary, EXP-05 had an error rate of 41.27% and EXP-08 had an error rate of 49.84%. Error increased with increasing distance from a boundary, with error rates at each distance group only 0.83% and 1.05%, respectively, for observations with >50 units of a boundary. EXP-05 had a lower error rate in each group, with its largest difference in the errors at two units from a boundary. These results indicate that classification was strongly related to the construction of the target.

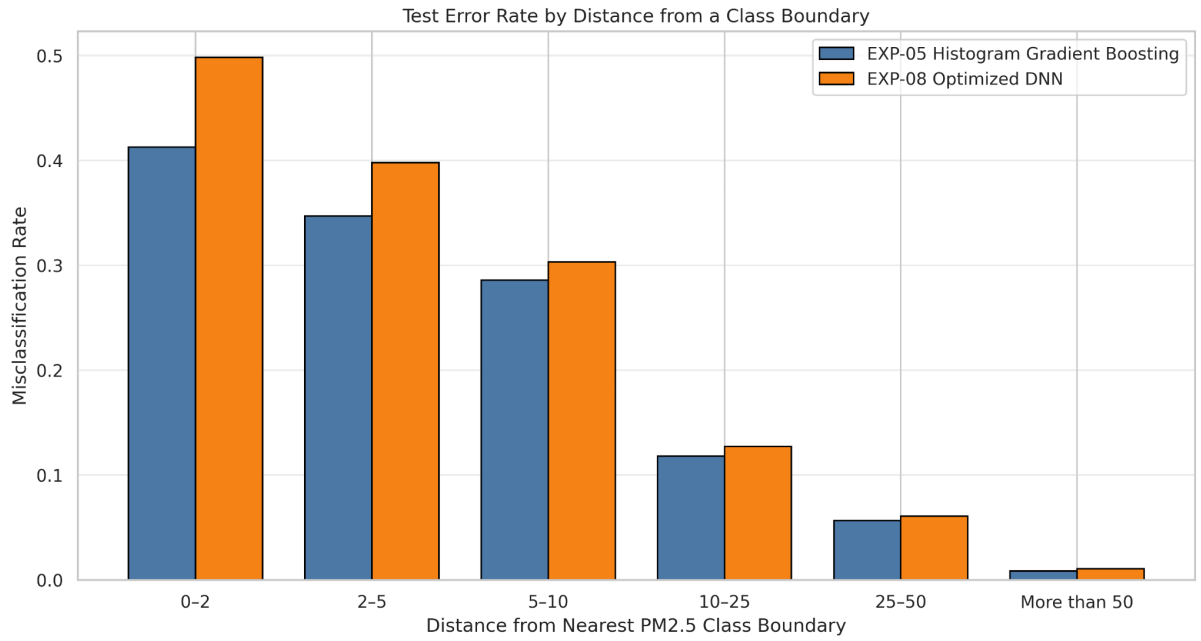


Fig. 4. Test misclassification rate by distance from the nearest PM2.5 class boundary.

V. DISCUSSION

The findings show that model complexity did not automatically produce better air-quality classification. Histogram Gradient Boosting achieved the strongest final test results, outperforming the optimized deep neural network in accuracy, macro-F1 and macro ROC-AUC. It also required considerably less training time. This suggests that boosting was particularly suitable for the structured numerical and categorical features used in this project. Similar studies have found that tree-based ensembles can perform strongly on air-quality data because they capture nonlinear relationships without requiring large neural architectures [8]–[10]. Class weighting improved balanced performance, although it sometimes reduced overall accuracy. This was particularly visible in the Logistic Regression experiments, where weighting increased macro recall and macro-F1 but reduced accuracy. The result demonstrates why accuracy should not be used alone with imbalanced targets. Selecting models using macro-F1 ensured that the less frequent Elevated and Very High categories received equal importance during evaluation. The Elevated category remained the most difficult class. Its lower precision indicates that many observations predicted as Elevated belonged to neighbouring categories. Boundary analysis supports this interpretation. EXP-05 produced an error rate of 41.27% for observations within two units of a PM2.5 class boundary, compared with only 0.83% for observations more than 50 units from a boundary. EXP-08 showed the same pattern, with corresponding error rates of 49.84% and 1.05%. Therefore, much of the remaining error occurred where small changes in PM2.5 produced different class labels, even though the underlying pollution conditions were similar. Permutation importance identified PM10, CO and dew point as the leading predictors. PM10 and CO are related to combustion, traffic and other pollution sources, while dew point represents atmospheric moisture conditions that can affect pollutant accumulation and dispersion. However, permutation importance measures predictive contribution rather than causation. The results should therefore not be interpreted as proof that directly changing one of these variables would cause a particular air-quality outcome. The learning curve showed that validation macro-F1 improved as more chronological training data were included, while the training–validation gap became smaller. This suggests manageable model variance and indicates that additional representative

observations could further improve performance. Nevertheless, the models classify conditions for the current hour rather than forecasting future pollution. Their immediate value is therefore in monitoring and decision support rather than advance warning. Several limitations should be considered. The severity groups were created from fixed PM2.5 thresholds, so observations near a boundary may be separated despite having similar concentrations. Hourly observations from the same stations are also dependent, and pollution relationships may change over time. Missing sensor values required imputation, and findings from Beijing cannot be transferred directly to African cities because emission sources, climate, infrastructure and monitoring coverage differ. Despite these limitations, the workflow provides a useful foundation for air-quality projects in Africa. A future implementation should be retrained and validated using local sensor data. It could then support pollution alerts, identify high-risk periods and help environmental authorities allocate monitoring resources. It should complement emission inventories and public-health evidence rather than be treated as a standalone pollution-reduction intervention.

VI. CONCLUSION

This project compared traditional machine-learning and deep-learning approaches for classifying hourly Beijing air-quality observations into four PM2.5 severity groups. Eight experiments were evaluated using chronological training, validation and test partitions to represent performance on future observations and reduce temporal leakage. Histogram Gradient Boosting achieved the strongest final performance, with an accuracy of 0.8580, macro-F1 of 0.8226 and macro ROC-AUC of 0.9731. The optimized deep neural network remained competitive but produced a lower macro-F1 of 0.8067 while requiring more training time. The findings demonstrate that a well-designed traditional ensemble can outperform a more computationally demanding neural network on structured environmental data. PM10, CO and dew point were the most influential predictors. The Elevated class remained the most difficult to identify, and errors were concentrated close to the PM2.5 thresholds used to define the target categories. This shows that some incorrect classifications reflected ambiguity around artificial class boundaries rather than an inability to distinguish broadly different pollution conditions. The completed workflow offers a reproducible foundation for future air-quality monitoring projects. Further research should evaluate forecasting models, ordinal classification, probability calibration and locally collected data from African cities. Such models could support earlier warnings and more targeted environmental decisions, but they should be combined with local health evidence, emission data and policy expertise when designing interventions to reduce air pollution.

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